Krimelack

Content-Addressable Structural Memory with
Clockless Habituation for Ternary Perception Systems

Version 1.0 — May 2026 — Confidential

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Abstract

We present Krimelack, a content-addressable structural memory architecture for balanced ternary perception systems. Krimelack stores *structural motifs* — settled states of a combinational coupling fabric — and provides cue-based recall with clockless habituation feedback. Unlike conventional content-addressable memories (CAMs) that match bit patterns, Krimelack stores coupled multi-field patterns in balanced ternary $(-1, 0, +1)$ and scores matches by counting agreeing non-null trits, giving the null state (0) a computational role: it represents structural uncertainty, not a value to be matched.

The architecture combines four capabilities that do not exist together in any prior system: (i) structural motif storage with quality-gated commits, (ii) parallel cue-based recall from partial signatures, (iii) pattern-level habituation via combinational dead-zone feedback, and (iv) autonomous commit gating based on dimensional exhaustion detection. Krimelack has been implemented in Verilog, validated on Xilinx Zynq-7020 FPGA silicon, and demonstrated in autonomous robot navigation with camera and IR sensor fusion (351 samples, 176 seconds continuous operation).

Gate count: ∼4,600 equivalent gates per block (64-entry, 48-bit). **Recall latency:** combinational (zero clock cycles). **Prior art:** none for the combined architecture.

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1 Introduction

1.1 The Problem

Perception systems that operate without training — where a single exposure must suffice for future recognition — require a memory architecture fundamentally different from neural network weight matrices or conventional lookup tables.

The requirements are:

1. **Store structural patterns, not raw data.** The memory must capture the settled state of a perception system, not the sensor input that caused it.
2. **Recall from partial cues.** A noisy or incomplete sensor input must retrieve the closest stored pattern, not fail on exact mismatch.
3. **Reject noise.** Patterns captured during ambiguous or uncertain perception must be blocked from storage.
4. **Habituate to familiarity.** Repeated exposure to the same environment must not produce the same response indefinitely. The system must seek novelty.
5. **Operate without a clock in the recall path.** For integration with combinational perception hardware (nanosecond settling), recall must be combinational.

No existing memory architecture satisfies all five requirements.

1.2 Context: The SPPU Perception Fabric

Krimelack is designed for the ArcLoom Structural Perception Processing Unit (SPPU), a combinational balanced ternary coupling fabric that settles to perception decisions through wire propagation in $\sim$5 ns. The SPPU produces a *loom state*: a vector of balanced ternary trits $(-1, 0, +1)$ encoded as 2-bit pairs across multiple coupled strands.

The loom state captures what the perception system *committed to* given a set of sensor inputs. Trits at ± 1 represent structural commitments. Trits at 0 represent dimensions where coupling pressure was insufficient to force a decision — structural uncertainty, not missing data.

Krimelack stores these loom states as *motifs* and provides cue-based recall to enable temporal pattern recognition across perception events.

1.3 Naming

"Krimelack" derives from the concept of a "crime of the lacquer" — a structural record that cannot be erased once committed, like a flaw sealed under a coat of lacquer. A committed motif is a permanent structural fact, not a mutable variable.

2 Architecture

2.1 Memory Structure

Definition 2.1 (Motif)**.** *A **motif** is a stored loom state: a vector of N balanced ternary trits encoded as $2N$ bits.*

- *`00` = null (0): structural uncertainty*
- *`01` = positive (+1): excitatory commitment*
- *`10` = negative (−1): inhibitory commitment*
- *`11` = invalid (never stored)*

The current FPGA implementation uses:

Parameter	**Value**
Trits per motif	24 (8 strands $\times$ 3 settling/decision trits)
Bits per motif	48
Motif depth	32 entries (circular buffer)
Hash width	8 bits (XOR-fold duplicate detection)

2.2 Functional Blocks

Krimelack consists of four functional blocks:

1. **Motif Store:** A register array holding D motifs of W bits each, with a circular write pointer and occupancy counter.
2. **Commit Gate:** Quality filter that determines whether a candidate motif is eligible for storage. Prevents committing noise, duplicates, or ambiguous states.
3. **Recall Engine:** Parallel comparator that scores every stored motif against a query pattern simultaneously. Returns the best match and its score. Purely combinational — zero clock cycles for recall.
4. **Habituation Feedback:** The match score feeds back into the SPPU's dead-zone thresholds, raising the commitment barrier for familiar patterns. This is the mechanism by which Krimelack produces novelty-seeking behavior without any clock or timer.

3 Commit Gating

3.1 The Problem of Noisy Commits

A perception system that stores every loom state would fill its memory with noise — transient states, sensor glitches, and ambiguous readings. Krimelack uses *commit gating* to ensure that only structurally significant states are stored.

3.2 Commit Eligibility Criteria

A motif is committed if and only if ALL of the following hold:

1. **Structural lock (L6):** The dimensional exhaustion detector has fired, indicating that the loom state has collapsed to a geometrically inevitable configuration. Formally: the number of non-null trits exceeds the threshold n_{start}/e, where n_{start} is the total number of trits and $e = 2.718\ldots$ is Euler's number.
2. **Low uncertainty:** The adjusted uncertainty $U^* \leq U_{settle}$ (default: 0.35 in Q16.16 fixed-point). This ensures the perception system is confident about the current state.
3. **Sufficient resonance:** The resonance score $R \geq R_{commit}$ (default: 0.72 in Q16.16). This ensures the state has structural coherence, not just low noise.
4. **SafeMode inactive:** The system is not in a protective mode that suppresses storage.
5. **No duplicate:** The candidate motif does not already exist in memory. Checked via hash-based comparison (XOR-fold to 8 bits, followed by exact bit comparison on hash collision).

3.3 Forced Commit (Software Capture)

For calibration and training scenarios (e.g., turntable object capture), software can trigger a commit via AXI register write. The forced commit bypasses criteria (1)–(4) but still enforces criterion (5) — duplicate detection is never disabled. This ensures idempotency: presenting the same stimulus twice never produces two motifs.

3.4 Formal Specification

$$\text{commit} = \begin{cases} 1 & \text{if } (\text{L6\_fire} \wedge U^* \leq U_s \wedge R \geq R_c \wedge \neg\text{safe} \wedge \neg\text{dup}) \\ & \quad \vee\ (\text{force} \wedge \neg\text{dup}) \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

4 Recall: Parallel Pattern Matching

4.1 Scoring Function

The recall engine compares a query pattern Q against every stored motif M_k ($k = 0, \ldots, D-1$) simultaneously. The score counts the number of trits where both the query and the motif are non-null and agree:

$$\text{score}(Q, M_k) = \sum_{i=0}^{N-1} \begin{cases} 1 & \text{if } Q_i \neq 0 \wedge M_{k,i} \neq 0 \wedge Q_i = M_{k,i} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

4.2 The Role of the Null Trit

This scoring function treats the null trit (0) differently from +1 and −1:

- A null trit in the query means "I don't have information about this dimension." It neither contributes to nor penalizes the score.
- A null trit in the motif means "the perception system was uncertain about this dimension when the motif was committed." It neither contributes to nor penalizes the score.
- Only trits where *both* query and motif are committed (±1) and *agree* contribute to the score.

This is fundamentally different from binary CAM matching, where every bit contributes. In Krimelack, the null trit is *not* a value — it is the absence of commitment. Two patterns can match perfectly on all committed dimensions while having different numbers of null trits. This is correct behavior: if the perception system was uncertain about a dimension, that dimension should not affect recognition.

4.3 Best-Match Selection

The motif with the highest score is returned as the best match. Ties are broken by index (earliest stored motif wins — temporal priority).

$$k^* = \arg \max_{k:\, k<D,\, \text{count}[k]>0} \text{score}(Q, M_k) \tag{3}$$

The recall outputs are:

- `best_match`: the full motif M_{k^*}
- `match_score`: $\text{score}(Q, M_{k^*})$ as an 8-bit integer
- `recall_valid`: 1 if any match was found, 0 if memory is empty

4.4 Combinational Implementation

The entire recall engine is implemented as combinational logic. No clock is required. The query propagates through the comparator array and the best-match result appears at the output wires within one propagation delay.

For a 32-entry, 48-bit implementation:

- 32 parallel comparators (× 24 trits each)
- 32 score accumulators (8-bit each)
- 5-stage binary max-tree to find the best score
- Total: ∼2,000 equivalent gates

Recall latency is determined by propagation delay through the comparator tree, approximately 3–5 ns in 7nm FinFET. This matches the SPPU settling time, allowing the recall result to feed back into the same perception cycle.

5 Clockless Habituation

5.1 The Corner-Trapping Problem

A combinational perception system with a fixed dead zone produces identical output for identical input. If the system encounters a corner (a sensor state that produces a strong coupling response), it will commit to the same decision indefinitely. There is no escape. No clock-based timeout. No iteration counter. The system is trapped by its own correct behavior.

Traditional solutions require clocks or timers. Krimelack solves this without either.

5.2 Dead-Zone Feedback Mechanism

The match score from recall feeds directly into the SPPU's dead-zone threshold:

$$\boxed{\tau_{\text{eff}} = \tau_{\text{base}} + f(\text{match\_score})} \tag{4}$$

where f is a damped mapping (score changes must exceed a barrier of $\sim$15 units to propagate, preventing oscillation from small score fluctuations).

Familiar states (high match score) raise the commitment barrier. Trits that barely exceeded the dead zone now fall below it and revert to null. The loom state changes. A different decision emerges.

Novel states (low match score) leave the dead zone at its baseline. Full sensitivity. Maximum coupling.

No clock. No timer. No iteration. The feedback is a combinational loop: SPPU output $\rightarrow$ Krimelack recall $\rightarrow$ match score $\rightarrow$ SPPU dead zone $\rightarrow$ different SPPU output. The loop settles to a fixed point where the dead-zone adjustment is self-consistent.

5.3 Cognitive Consequences

The dead-zone feedback produces emergent behaviors that mirror biological cognition:

Behavior	Mechanism
Novelty seeking	Familiar states are harder to commit to; system gravitates toward unfamiliar states
Attention	Novel stimuli get full sensitivity; repeated stimuli get diminished sensitivity
Boredom	Sustained identical input raises dead zone until no trits commit; system goes silent
Creativity	Familiar responses suppressed; only novel decisions remain available
Recognition	Partial cue retrieves full motif; system "remembers" the structural state

These behaviors are not designed or programmed. They are mathematical consequences of dead-zone feedback from a content-addressable memory. They emerge from the architecture.

5.4 Stability Analysis

The feedback loop is stable because:

1. The damping barrier prevents small fluctuations from propagating (hysteresis)
2. The dead-zone increase is monotonic with match score (no oscillation)
3. The maximum dead-zone increase is bounded (cannot exceed the maximum coupling pressure, which would prevent all commitment)
4. The loop settles in one propagation delay (no multi-cycle ringing)

6 TSAC Native Habituation (Future)

On ferroelectric substrates (e.g., HfO_2-based Ternary State Analog Cells), habituation becomes a free physical property of the material.

6.1 Mechanism

Sustained polarization in one direction causes dielectric fatigue — the potential barrier rises with repeated identical state:

- Sustained +1 → barrier: 20 → 22 → 25 → 28
- Barrier exceeds coupling pressure → trit falls to null
- Null (rest) → barrier recovers → fresh sensitivity

The RC time constant of the ferroelectric material IS the habituation rate. No circuit. No logic. Physics.

6.2 Two-Layer Architecture

On ASIC, both layers operate simultaneously:

- **Substrate layer** (TSAC fatigue): fast, local, per-trit. Physical depletion of the commitment mechanism.
- **Cognitive layer** (Krimelack feedback): slower, global, pattern-level. Experiential familiarity across the full loom state.

This mirrors biology: receptor saturation (fast, local) and cortical habituation (slow, global) operating at different scales.

7 Gate Count and Physical Implementation

7.1 CMOS Gate Count

Component	Equivalent Gates
Motif register array (32 entries × 48 bits)	1,536
Hash table (32 entries × 8 bits)	256
Duplicate detection (32 parallel hash + exact compare)	200
Commit control (eligibility logic + write pointer)	100
Recall comparators (32 × 24-trit parallel match)	1,600
Score accumulators (32 × 8-bit)	400
Best-match tree (5-stage binary max)	200
Habituation feedback (match score → dead-zone adder)	144
Total per Krimelack block	**∼4,600**

7.2 Scaling

Configuration	Entries	Gates	Recall (7nm)
Minimal (FPGA demo)	32	4,600	∼3 ns
Standard (ArcLoom-256)	64	8,400	∼4 ns
Extended (vision)	256	30,000	∼6 ns

Recall latency scales logarithmically with depth (max-tree depth = $\log_2 D$). Even at 256 entries, recall completes within the SPPU settling window.

7.3 Power

Krimelack's recall path is combinational: zero dynamic power between queries. The commit path requires a clock edge for register writes, but commits are rare (only when L6 fires and all eligibility criteria are met). Typical commit rate: <1 Hz. Power contribution: negligible.

8 Comparison with Prior Art

8.1 Ternary Content-Addressable Memory (TCAM)

TCAMs are deployed in every enterprise network router. They match against three states (0, 1, X = don't care) in parallel.

Property	TCAM	Krimelack
Third state meaning	Wildcard (don't care)	Structural uncertainty
Match scoring	Exact match or fail	Graded (count of agreements)
Null contribution	Matches anything	Contributes nothing
Habituation	None	Dead-zone feedback
Commit gating	None	Quality-filtered
Application	Address lookup	Structural perception

8.2 Hopfield Networks

Hopfield networks store patterns as energy minima and recall via iterative energy minimization.

Property	Hopfield	Krimelack
Storage method	Distributed in weights	Explicit motif registers
Recall method	Iterative settling	Single combinational pass
Weight symmetry	Required ($W_{ij} = W_{ji}$)	Not applicable
Training	Hebbian learning rule	None (commit gating)
Capacity	$\sim 0.14N$ patterns	D entries (configurable)
Spurious attractors	Yes (mixture states)	No (exact storage)
Recall time	Multiple iterations	∼3–5 ns

8.3 Memristor Habituation (Nature Communications, 2025)

Third-order HfO_2/TiO_x memristors with physical habituation. Robot arm ignores 71% of familiar stimuli.

Property	Memristor	Krimelack
Habituation level	Per-synapse (local)	Pattern-level (global)
Habituation type	Analog (continuous)	Digital (threshold)
Context	Embedded in trained SNN	Standalone, no ML
Recall type	SNN inference	Combinational CAM
Training required	Yes (SNN weights)	No

8.4 Summary: What Krimelack Has That Nobody Else Has

1. **Structural motif storage** — coupled multi-field ternary patterns, not bit vectors or weight distributions

2. **Null-trit-aware scoring** — uncertainty dimensions excluded from matching, not wildcarded
3. **Quality-gated commits** — dimensional exhaustion + uncertainty + resonance filtering prevents noise storage
4. **Clockless pattern-level habituation** — dead-zone feedback from match score, no clock, no timer
5. **Combined in one architecture** — no prior system provides all four

9 Validation

9.1 FPGA Implementation

Krimelack is implemented in Verilog (`arcloom_krimelack.v`, 167 lines) and validated on Xilinx Zynq-7020 (PYNQ-Z2 board).

9.2 Demonstrated Capabilities

Capability	**Status**	**Evidence**
Motif commit gating (L6)	**Proven**	Motifs stored only on structural lock
Forced commit (AXI)	**Proven**	Software-triggered capture via register write
Duplicate detection	**Proven**	Same stimulus never produces two motifs
Cue-based recall	**Proven**	Partial query retrieves best match
Dead-zone feedback	**Proven**	Familiarity damping in live sensor loop
Autonomous navigation	**Proven**	351 samples, 176s, 3 IR + camera
Vision discrimination	**Proven (sim)**	5 objects, 0.11–0.36 match gaps
One-shot learning	**Proven**	Single turntable capture → recognition

9.3 Vision Results

Five realistic objects discriminated using camera-derived motifs:

- Match gap between objects: 0.11–0.36 (robust discrimination)
- Same object with different noise: 0.895–0.954 (reliable recall)
- Memory per object: ~4 KB structural fingerprint
- Training: zero. Single exposure via forced commit.
- Hardware: ~4,600 gates of Krimelack + ~2,250 gates of SPPU = <7,000 gates total

10 Applications

10.1 Immediate (FPGA)

- Autonomous robot obstacle memory ("I've been here before")
- Object recognition from turntable capture (one-shot learning)
- Sensor anomaly detection (novel states vs familiar baseline)

10.2 Near-Term (ASIC)

- Implantable medical sensors: cardiac rhythm motifs stored on-chip, anomaly = novel pattern not matching any stored motif
- Environmental monitoring: decade-life sensors that learn their “normal” environment and alert on structural novelty
- Safety-critical systems: deterministic pattern recognition with provable behavior (combinational, no software)

10.3 Long-Term (TSAC)

- Ferroelectric substrate: per-trit habituation as a free physical property
- Two-layer habituation: substrate (fast, local) + Krimelack (slow, global)
- Brain-scale structural memory: millions of motifs with physics-based adaptation

11 Conclusion

Krimelack is a content-addressable structural memory architecture that stores balanced ternary motifs with quality-gated commits, parallel cue-based recall, and clockless pattern-level habituation. It is the first memory system to combine all four capabilities in a single architecture.

The null trit is not a wildcard or a don’t-care bit. It is structural uncertainty — a first-class computational state that influences matching, habituation, and dimensional exhaustion detection. This distinction is fundamental: Krimelack does not *match patterns*. It recognizes structural states by measuring how many committed dimensions agree. Dimensions where the system was uncertain are honestly excluded.

The habituation mechanism — dead-zone feedback from match score — produces novelty-seeking, attention, boredom, and creativity as emergent consequences of the architecture, not as designed behaviors. These cognitive properties arise from the mathematics of a combinational feedback loop between a content-addressable memory and a coupling fabric with dead zones.

Krimelack has been validated on FPGA silicon in autonomous robot navigation with camera and IR sensor fusion. It operates at ∼4,600 gates with combinational recall latency of ∼3–5 ns. No prior art exists for the combined architecture.
